

FREEDOM INTERNATIONAL SCHOOL

WORKSHEET (MCQ) - SOLUTIONS

PHYSICS

ELECTROSTATIC POTENTIAL AND CAPACITANCE

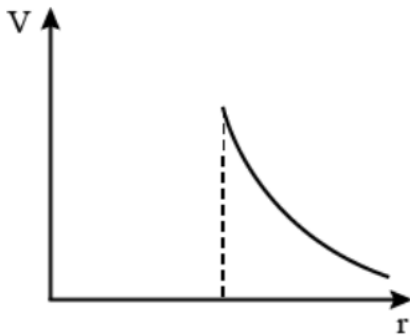
1. The electric potential at the surface of an atomic nucleus ($Z=50$) of radius 9.0×10^{-13} cm is
(a) 9×10^5 V (b) 8×10^6 V (c) 80 V (d) 9 V

Ans: (b) 8×10^6 V

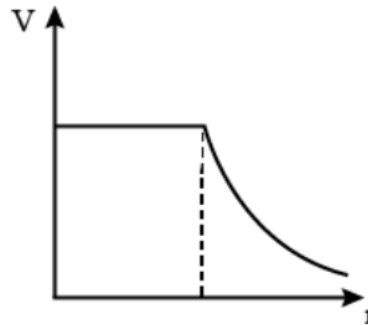
$$\begin{aligned}\text{Electric potential, } V &= \frac{1}{4\pi\epsilon_0} \frac{Ze}{r} \\ &= \frac{9 \times 10^9 \times 50 \times 1.6 \times 10^{-19}}{9 \times 10^{-13} \times 10^{-2}} \\ &= 8 \times 10^6 \text{ V}\end{aligned}$$

2. In the case of a charged metallic sphere, potential (V) changes with respect to distance (r) from the centre as

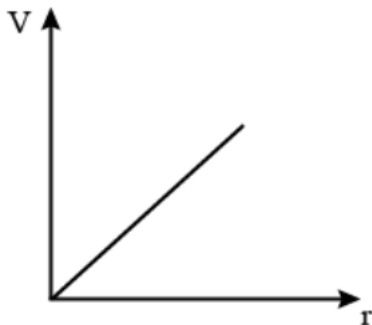
(a)



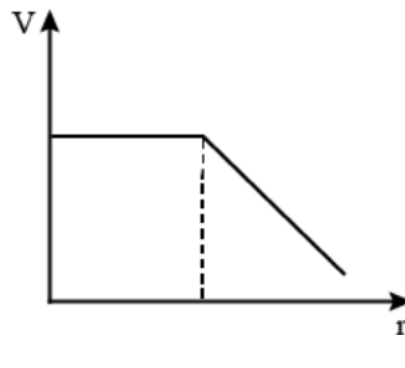
(b)



(c)



(d)



Ans: (b)

Potential at any point inside the sphere is constant and is equal to that on the surface.

At outside points, $V \propto 1/r$.

3. In bringing an electron towards another electron, the electrostatic potential energy of the system
 (a) increases (b) decreases (c) remains unchanged (d) becomes zero

Answer: (a) increases

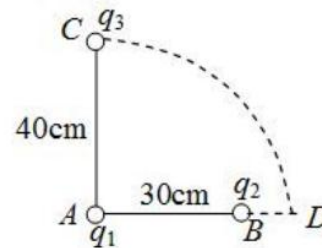
In bringing an electron towards another electron work has to be done since same charges repel each other. The work done stored as electrostatic potential energy and hence electrostatic potential energy of the system increases.

4. The potential on the hollow sphere of radius 1 m is 1000 V, then potential at $\frac{1}{4}$ m from the centre of the sphere is
 (a) 1000 V (b) 500 V (c) 250 V (d) zero V

Answer: (a) 1000 V

The potential at any point inside the charged hollow metallic sphere is same as that on its surface.

5. Two charges q_1 and q_2 are placed 30 cm apart as shown in the figure. A third charge q_3 is moved along the arc of a circle of radius 40 cm from C to D. The change in the potential energy of the system is $\frac{q_3 k}{4\pi\epsilon_0}$, where k is



- (a) $8 q_2$ (b) $8 q_1$ (c) $6 q_2$ (d) $6 q_1$

Ans: (a) $8 q_2$

The potential energy when q_3 is at point C

$$U_1 = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_3}{0.40} + \frac{q_2 q_3}{\sqrt{(0.40)^2 + (0.30)^2}} \right]$$

The potential energy when q_3 is at point D

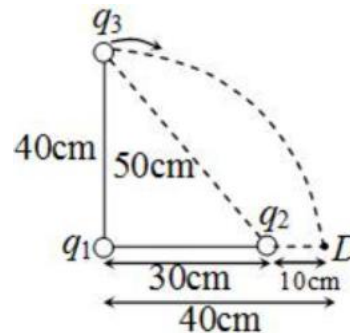
$$U_2 = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_3}{0.40} + \frac{q_2 q_3}{0.10} \right]$$

$$\Delta U = U_2 - U_1$$

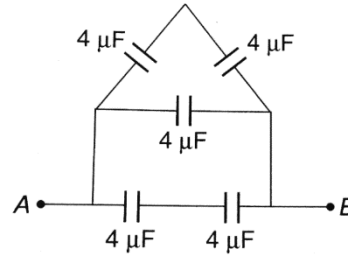
$$= \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_3}{0.40} + \frac{q_2 q_3}{0.10} - \frac{q_1 q_3}{0.40} - \frac{q_2 q_3}{0.50} \right]$$

$$\Delta U = \frac{1}{4\pi\epsilon_0} [8q_2 q_3] = \frac{q_3}{4\pi\epsilon_0} (8q_2)$$

$$\therefore k = 8q_2$$

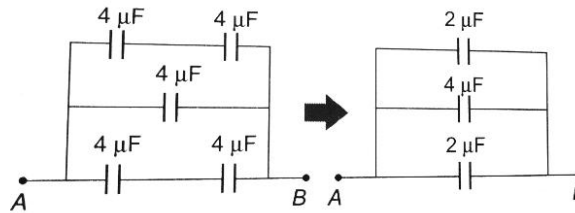


6. Equivalent capacitance between A and B is



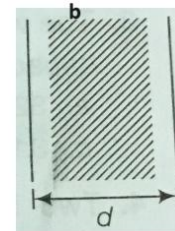
- (a) $8 \mu\text{F}$ (b) $6 \mu\text{F}$ (c) $268 \mu\text{F}$ (d) $10/38 \mu\text{F}$

Ans: (a) $8 \mu\text{F}$



$$C_{AB} = 8\mu\text{F}$$

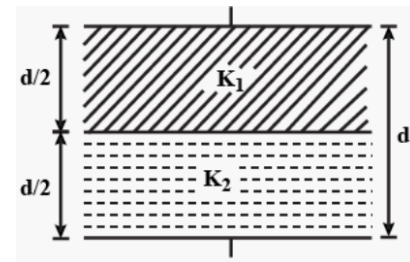
7. A copper plate of thickness b is placed inside a parallel capacitor of plate distance d and area A as shown in the figure. The capacitance of the capacitor is



- (a) $\frac{A\epsilon_0}{d}$ (b) $\frac{A\epsilon_0}{b}$ (c) $\frac{A\epsilon_0}{d-b}$ (d) infinity

Ans: (c) $\frac{A\epsilon_0}{d-b}$

8. A parallel plate capacitor of plate area A and separation d is filled with dielectric as shown in the figure. The dielectric constants are K_1 and K_2 . Net capacitance is



- (a) $\frac{\epsilon_0 A}{d} (k_1 + k_2)$ (b) $\frac{\epsilon_0 A}{d} \left(\frac{k_1 + k_2}{k_1 k_2} \right)$ (c) $\frac{2\epsilon_0 A}{d} \left(\frac{k_1 k_2}{k_1 + k_2} \right)$ (d) $\frac{2\epsilon_0 A}{d} \left(\frac{k_1 + k_2}{k_1 k_2} \right)$

Ans: (c) $\frac{2\epsilon_0 A}{d} \left(\frac{k_1 k_2}{k_1 + k_2} \right)$

The arrangement is equivalent to series combination of two capacitors of plate area A and separation $d/2$
 $1/C = 1/C_1 + 1/C_2$

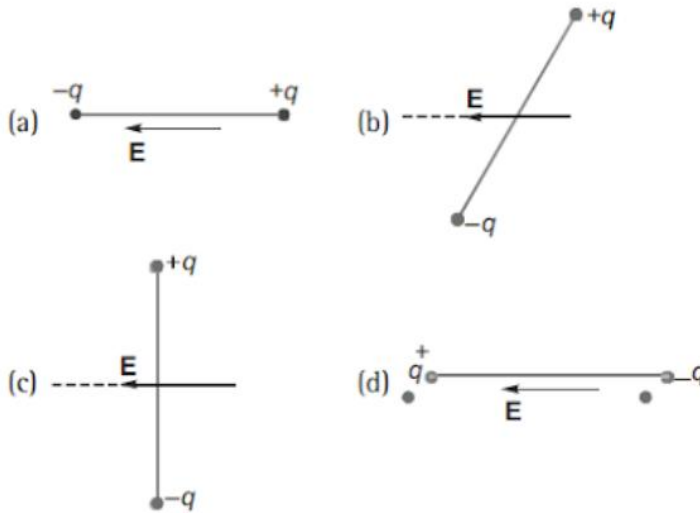
$$C_1 = \frac{\epsilon_0 A k_1}{d/2} \quad C_2 = \frac{\epsilon_0 A k_2}{d/2}$$

$$\therefore \frac{1}{C} = \frac{d}{2K_1\epsilon_0 A} + \frac{d}{2K_2\epsilon_0 A}$$

$$= \frac{d}{2\epsilon_0 A} \left(\frac{1}{K_1} + \frac{1}{K_2} \right) = \frac{d}{2\epsilon_0 A} \frac{K_1 + K_2}{K_1 K_2}$$

or $C = \frac{2\epsilon_0 A}{d} \frac{K_1 K_2}{K_1 + K_2}$

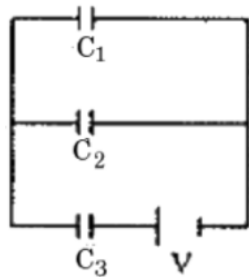
9. In which of the states shown in the figure, is potential energy of an electric dipole maximum?



Ans: (a)

Dipole moment vector is antiparallel to electric field vector.

10. Find the voltage across C_3 .



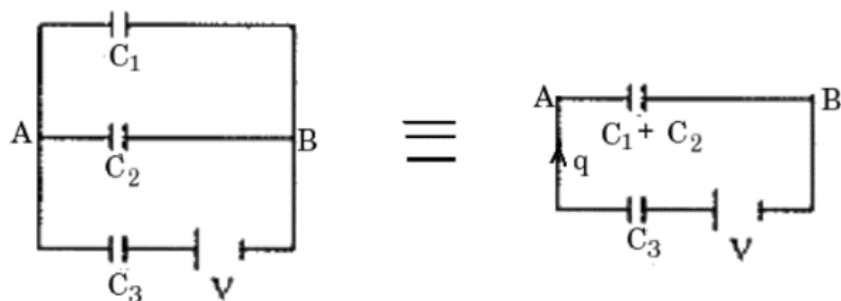
(a) $\frac{(c_1+c_2)V}{c_1+c_2+c_3}$

(b) $\frac{c_1V}{c_1+c_2+c_3}$

(c) $\frac{c_2V}{c_1+c_2+c_3}$

(d) $\frac{c_3V}{c_1+c_2+c_3}$

Ans: (a) $\frac{(c_1+c_2)V}{c_1+c_2+c_3}$



Capacitors C_1 and C_2 are in parallel, their equivalent capacitance is in series with capacitor C_3 . Hence, the equivalent capacitance of the given circuit is

$$C_{eq} = \frac{(C_1 + C_2)C_3}{C_1 + C_2 + C_3}$$

Charge on capacitor C_3 is

$$Q = V C_{eq} = \frac{V(C_1 + C_2)C_3}{C_1 + C_2 + C_3}$$

Voltage across C_3 is

$$= \frac{Q}{C_3} = \frac{V(C_1 + C_2)}{C_1 + C_2 + C_3}$$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- C. Assertion is true but Reason is false.
- D. Both Assertion and Reason are false.

11. **Assertion:** Increasing the charge on the plates of a capacitor means increasing the capacitance.
Reason: Capacitance is directly proportional to charge.
 Ans: (D)
12. **Assertion:** Dielectric polarization means formation of positive and negative charges inside the dielectric.
Reason: Free electrons are formed in this process.
 Ans: (D)
13. **Assertion:** When air between the plates of a parallel plate condenser is replaced by an insulating medium of dielectric constant its capacity increases.
Reason: Electric field intensity between the plates with dielectric in between it is reduced.
 Ans: (A)
14. **Assertion:** The electrostatic force between the plates of a charged isolated capacitor decreases when dielectric fills whole space between plates.
Reason: The electric field between the plates of a charged isolated capacitance increases when dielectric fills whole space between plates.
 Ans: (C)
15. **Assertion:** In the absence of an external electric field, the dipole moment per unit volume of a polar dielectric is zero.
Reason: The dipoles of a polar dielectric are randomly oriented.
 Ans: (A)